Pranay Tankala

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I am a Ph.D. student at Harvard advised by Cynthia Dwork. My interests lie in theoretical computer science, especially questions about artificial intelligence related to pseudorandomness, privacy, fairness, and alignment.

Education

Harvard University — Cambridge, MA
Ph.D. in Computer Science — Expected May 2027
Advisor: *Cynthia Dwork*
Dissertation: *Structure and Randomness in Artificial Intelligence*

S.M. in Computer Science — 2020 – 2021

A.B. in Computer Science and Mathematics, Secondary in Statistics — 2017 – 2021
summa cum laude

Cary Academy — Cary, NC
Independent secondary school — 2010 – 2017

Employment

Apple · Machine Learning Research (MLR) — Cupertino, CA
Intern with Parikshit Gopalan, Vitaly Feldman, and Kunal Talwar — 2025, 2026

Honors

Best Paper Award — 2026
Symposium on Foundations of Responsible Computing (FORC)

Certificate of Distinction in Teaching — 2018, 2019, 2022
Awarded by Derek Bok Center, Harvard University

International Collegiate Programming Contest
2nd place among North American teams at the 2019 World Finals — Porto, Portugal
3rd place team at the 2020 North American Championship — Atlanta, GA

Phi Beta Kappa · Junior 24 — 2020
First cohort of inductees from Harvard College Class of 2021

John Harvard Scholar — 2019, 2020
Top 5% of class at Harvard College

Detur Book Prize — 2018
High academic standing among first-year students at Harvard College

Publications

Agnostic Learning by Imitating 2026
with Sílvia Casacuberta, Parikshit Gopalan, Varun Kanade, Omer Reingold and Konstantinos Stavropoulos (preprint)

Efficient and Private Property Testing via Indistinguishability 2026
with Cynthia Dwork
Symposium on Foundations of Computer Science (FOCS)

Universal Hardness From Masking Channels 2026
with Parikshit Gopalan and Manoj Prabhakaran (preprint)

Efficient Calibration for Decision Making 2026
with Parikshit Gopalan, Konstantinos Stavropoulos, and Kunal Talwar
Symposium on Theory of Computing (STOC)

The Importance of Being Smoothly Calibrated 2026
with Parikshit Gopalan, Konstantinos Stavropoulos, and Kunal Talwar
Symposium on Foundations of Responsible Computing (FORC) ★ **Best Paper Award**

Differentially Private Verification of Distribution Properties 2026
with Elbert Du, Cynthia Dwork, and Linjun Zhang (preprint)

Differentially Private Learning Beyond the Classical Dimensionality Regime 2025
with Cynthia Dwork and Linjun Zhang
Theory of Cryptography Conference (TCC)

From Fairness to Infinity: Outcome-Indistinguishable (Omni)Prediction in Evolving Graphs 2025
with Cynthia Dwork, Chris Hays, Nicole Immorlica, and Juan C. Perdomo
Conference on Learning Theory (COLT)

Privately Estimating a Gaussian: Efficient, Robust, and Optimal 2023
with Daniel Alabi, Pravesh K. Kothari, Prayaag Venkat, and Fred Zhang
Symposium on Theory of Computing (STOC)

From Pseudorandomness to Multi-Group Fairness and Back 2023
with Cynthia Dwork, Daniel Lee, and Huijia (Rachel) Lin
Conference on Learning Theory (COLT)

K-Deep Simplex: Manifold Learning via Local Dictionaries 2023
with Abiy Tasissa, James Murphy, and Demba Ba
Transactions on Signal Processing (TSP)

Weighed ℓ1 on the Simplex: Compressive Sensing Meets Locality 2021
with Abiy Tasissa and Demba Ba
Statistical Signal Processing Workshop (SSP)

Invited Talks

Stanford "Recent Advances in Regularity Lemmas" Workshop — Stanford, CA
"Agnostic Learning by Imitating" — November 2026

Apple Machine Learning Research Seminar — Cupertino, CA
"Agnostic Learning by Imitating" — August 2026

Apple EXPLOR Team Research Meeting — Cupertino, CA
"Agnostic Learning by Imitating" — August 2026
"The Importance of Being Smoothly Calibrated" — June 2026
"Efficient Calibration for Decision Making" — August 2025
"Supersimulators" — July 2025

Harvard Privacy Tools Seminar — Boston, MA
"Differentially Private Learning Beyond the Classical Dimensionality Regime" — May 2025

Boston University & US Census Bureau Cooperative Agreement Fall Retreat — Boston, MA
"Privately Estimating a Gaussian: Efficient, Robust, and Optimal" — September 2023

Harvard "TGINF" Graduate Student Theory Seminar — Boston, MA
"Private High-Dimensional Statistics" — September 2023

Harvard Privacy Tools Seminar — Boston, MA
"Privately Estimating a Gaussian: Efficient, Robust, and Optimal" — April 2023

Conference Presentations

Symposium on Foundations of Computer Science (FOCS) — New York, NY
"Efficient and Private Property Testing via Indistinguishability" — November 2026

Symposium on Foundations of Responsible Computing (FORC) — Boston, MA
"The Importance of Being Smoothly Calibrated" — June 2026

Theory of Cryptography Conference (TCC) — Aarhus, Denmark
"Differentially Private Learning Beyond the Classical Dimensionality Regime" — December 2025

Symposium on Foundations of Responsible Computing (FORC) — Stanford, CA
"Differentially Private Learning Beyond the Classical Dimensionality Regime" — June 2025
Non-Archival Highlights Track

Theory and Practice of Differential Privacy Workshop (TPDP) — Mountain View, CA
"Differentially Private Learning Beyond the Classical Dimensionality Regime" — June 2025
Selected Oral Presentation

Conference on Learning Theory (COLT) — Bangalore, India
"From Pseudorandomness to Multi-Group Fairness and Back" — July 2023

Teaching

Substitute Lecturer April 2026
Prof. Cynthia Dwork, Harvard CS 2260
Topics in Theory for Society: Differential Privacy

Guest Lecturer March 2024
Prof. Cynthia Dwork, Harvard CS 226
Topics in Theory for Society: Fairness and Validity
"From Pseudorandomness to Multi-Group Fairness and Back"

Guest Lecturer March 2023
Prof. Cynthia Dwork, Harvard CS 226
Topics in Theory for Society: Differential Privacy
"Private High-Dimensional Statistics"

Teaching Fellow January 2022 – May 2022
Prof. Cynthia Dwork, CS 226
Topics in Theory for Society: Fairness and Validity

Teaching Fellow August 2019 – December 2019
Prof. Boaz Barak, CS 121
Introduction to Theoretical Computer Science

Teaching Fellow January 2019 – May 2019
Prof. Jelani Nelson, CS 124
Data Structures and Algorithms

Teaching Fellow August 2018 – December 2018
Prof. Boaz Barak, CS 121
Introduction to Theoretical Computer Science

Service

Conference Reviewer

Innovations in Theoretical Computer Science (ITCS 2027)	October 2026
Symposium on Discrete Algorithms (SODA 2027)	August 2026
Symposium on Foundations of Computer Science (FOCS 2024)	May 2024
Conference on Algorithmic Learning Theory (ALT 2024)	November 2023
Symposium on Foundations of Computer Science (FOCS 2023)	May 2023

Pre-FORC Organizer May 2024
Organized two-day event for students and postdocs prior to the
Symposium on Foundations of Responsible Computing (FORC 2024)

Community Event Organizer May 2023, May 2024

Harvard Theory of Computation Group Spring Music Recital

Other Honors

North Carolina State Piano Competition (MTNA)

2nd Place, Junior Division 2013

Honorable Mention, Senior Division 2014, 2015

North Carolina All-State Honors Band

1st Chair Clarinet 2015, 2016